RS Vorlesung 7

HA) 1)

$$D = \frac{-\ln(M_p)}{\sqrt{\pi^2 + \ln^2(M_p)}} = 0,49$$

$$t_S = \frac{4}{D\omega_0} \Rightarrow \omega_0 = \frac{4}{t_S D} = 0,89$$

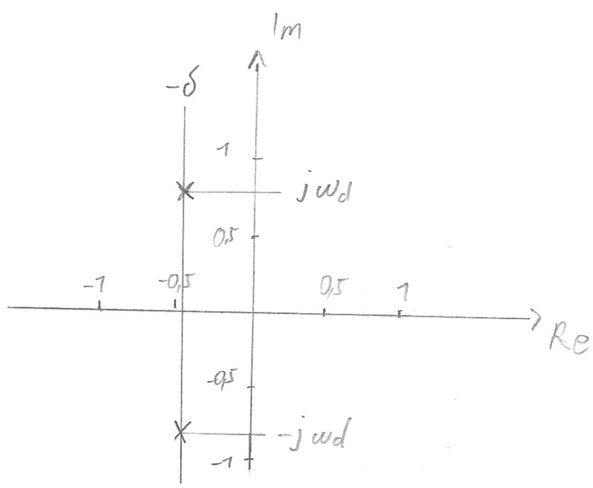
$$G(s) = \frac{\omega_0^2}{s^2 + 2D\omega_0 s + \omega_0^2} = \frac{0,78}{s^2 + 0,87s + 0,78}$$

$$p_{12} = -\omega_0 D \pm j\omega_0 \sqrt{1-D^2}$$

$$-0,44 \pm j \cdot 0,78 \qquad \leftarrow -\delta \pm j\omega_d$$

$\delta = 0,44$

$\omega_d = 0,78$



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HA) 2) $(33 + 15s + 3s^2)Y = \cancel{u(t)}\, U$

$$\frac{Y}{U} = \frac{1}{3(s^2+5s+11)} = \cancel{\frac{1}{s}\,\frac{1}{3}\sqrt{}}$$

$$\cancel{= \frac{1}{3\sqrt{}}} = \frac{1}{33} \quad \frac{11}{s^2+5s+11}$$

„muss + sein"

$$\omega_0 = \pm\sqrt{11} = \pm 3{,}32 \qquad t_s = \frac{4}{D\omega_0} = 1{,}61$$

$$D = \frac{\cancel{\omega_0}\,5}{2\omega_0} = 0{,}75$$

$$M_p = e^{-\frac{D\pi}{\sqrt{1-D^2}}} = 0{,}0284 = 2{,}84\%$$

$$t_p = \frac{\pi}{\omega_0\sqrt{1-D^2}} = 1{,}43$$

K = y∞/u0 = 10/2 = 5

HA) 3) a) ~~$M_p = 13{,}7$~~

$t_p = 33$

$M_p = 37\%\ (13{,}7)$

$$D = \frac{-\ln(M_p)}{\sqrt{\pi^2 + \ln^2(M_p)}} = 0{,}3 \qquad G_1(s) = \frac{\omega_0^2}{s^2 + 2D\omega_0 s + \omega_0^2}\,5$$

$$\omega_0 = \frac{\pi}{t_p\sqrt{1-D^2}} = 0{,}1 \qquad G_1(s) = \frac{10^{-2}}{s^2 + 6\cdot 10^{-2}s + 10^{-2}}\,5$$

b) $T_u = 21\,ms$

$T_g = 90 - 21 = 69\,ms$

$$\frac{T_g}{T_1} = \cancel{5{,}12}\ 4{,}46 \Rightarrow T_1 = \frac{T_g}{\cancel{5{,}12}\ 4{,}46} = \cancel{13{,}48}\ 15{,}47$$

$$T_1 = \frac{15{,}47 + 14{,}79}{2}$$

$$\frac{T_g}{T_u} = 3{,}29 \quad n = \cancel{7}$$

$$\frac{T_u}{T_1} = \cancel{2{,}1}\ 1{,}42 \Rightarrow T_1 = \frac{T_u}{\cancel{2{,}1}\ 1{,}42} = \cancel{10}\ 14{,}79 \qquad T_1 = 15{,}13$$

$$K = \frac{Y_\infty}{U_0} = 5$$

$$G_2(s) = \frac{5}{(1 + 15{,}13 s)^4}$$

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HA) 3) c)

$T_u = 3\,ms$

$T_g = 42\,ms$

$T_g = \frac{42}{2,72} = 15,44$

$T_1 = \frac{3}{0,28} = 10,71$

$T_1 = 13,08$

$\frac{T_g}{T_u} = \frac{42}{3} = 14 \qquad n = 2$

$G_3(s) = \frac{5}{(1+13,08)^2}$

```
clear all;
close all;
```

Hausaufgabe 3

```
% Definition
s = tf('s');

% Sprungantwort


% Übertragungsfunktionen
G1 = 5 * 0.01 / (s^2 + 0.06*s + 0.01);
G2 = 5 / (1 + 15.13*s)^4;
G3 = 5 / (1 + 13.08*s)^2;

% Plot
stepplot(2*G1,'r', 2*G2,'b', 2*G3,'g');
legend('G1(s)', 'G2(s)', 'G3(s)');
grid on;
```